

Simulation Results

Smart-Root-Dumb-Tree Model

Search scope : Combat move selection only - attacks only 1 territory in a round

Selection policy : UCB1 with exploration constant $c = 1.414$

Expansion : Single node per iteration

Simulation policy : Uniform random rollout for all players - keeps the tree dumb and root smart - excludes purchase and placement delegates - the simulation move generation has no pruning apart from time budget limit

Forward model : Handles the logs from the game, provides the requested move for the MCTS player (tree selects for combat and random for other delegates)

State evaluation : Territory strength ration + depth based penalties for non-terminal states

Playout : Fast playout by speeing legal move genration using game heuristics. The combat move generator gives 1 move with cheapest unit for undefended territory and a set of winnable moves with [min, max] strength for defended territories

Time budget : 1s per combat turn

Max tree depth : 10

Goal : Isolate and measure combat decision quality independent of the other game delegates

The reachability cheakcc is perfomed in Cpp and the linked to the python script using Pybind

Efficiency summary

MCTS Efficiency Summary (Game-wise)

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Game 1

Total rounds: 56

Iterations per round:

Mean: 6.32

Min : 5

Max : 14

Exploration_coverage (actions_explored / total_actions):

Mean: 0.1563

Min : 0.0000

Max : 1.0000

Avg Tree depth per round:

Mean: 9.4798

Min : 8.6667

Max : 10.0000

Game 2

Total rounds: 25

Iterations per round:

Mean: 7.92

Min : 6

Max : 22

Exploration_coverage (actions_explored / total_actions):

Mean: 0.2446

Min : 0.0000

Max : 1.0000

Avg Tree depth per round:

Mean: 9.5433

Min : 8.8571

Max : 10.0000

Game 3

Total rounds: 27

Iterations per round:

Mean: 7.41

Min : 6

Max : 21

Exploration_coverage (actions_explored / total_actions):

Mean: 0.2596

Min : 0.0000

Max : 1.0000

Avg Tree depth per round:

Mean: 9.5081

Min : 8.8333

Max : 10.0000

Game 4

Total rounds: 32

Iterations per round:

Mean: 7.06

Min : 5

Max : 21

Exploration_coverage (actions_explored / total_actions):

Mean: 0.2540

Min : 0.0000

Max : 1.0000

Avg Tree depth per round:

Mean: 9.3547

Min : 8.8333

Max : 10.0000

Game 5

Total rounds: 22

Iterations per round:

Mean: 7.36

Min : 6

Max : 13

Exploration_coverage (actions_explored / total_actions):

Mean: 0.2449

Min : 0.0001

Max : 1.0000

Avg Tree depth per round:

Mean: 9.5725

Min : 8.7143

Max : 10.0000

OVERALL AVERAGES (Across Games)

Iterations per round (game-averaged):

Mean: 7.21

Min: 5

Max: 22

Exploration coverage (game-averaged):

Mean: 0.23

Min: 0.00

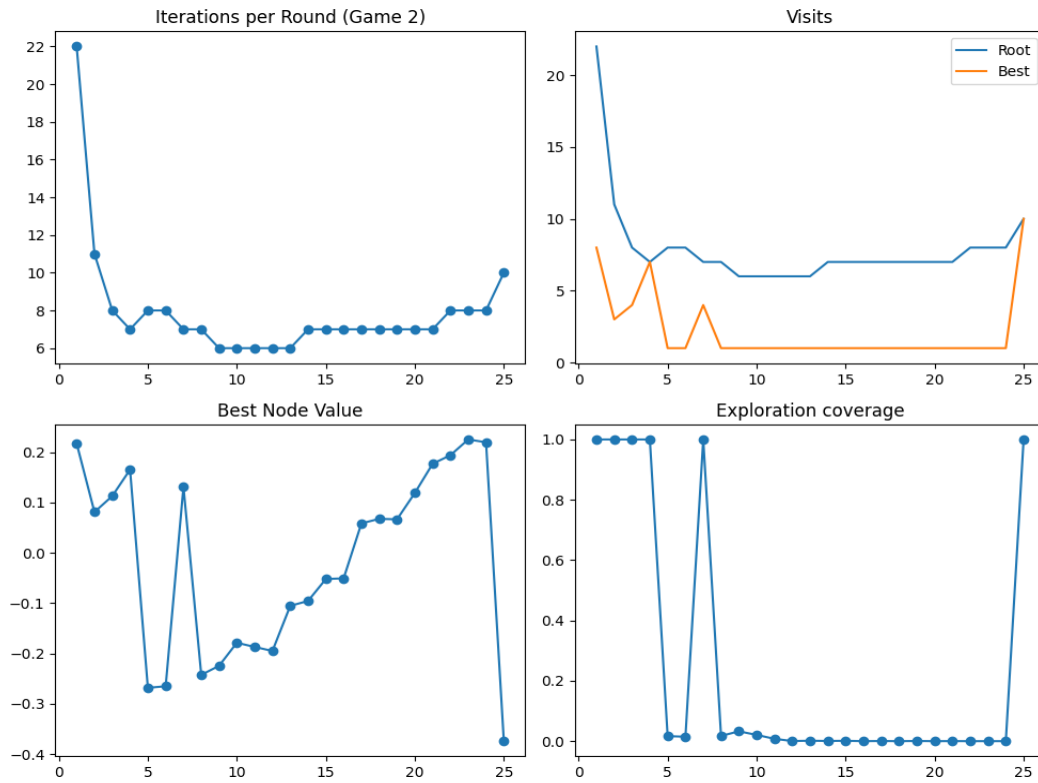
Max: 1.0

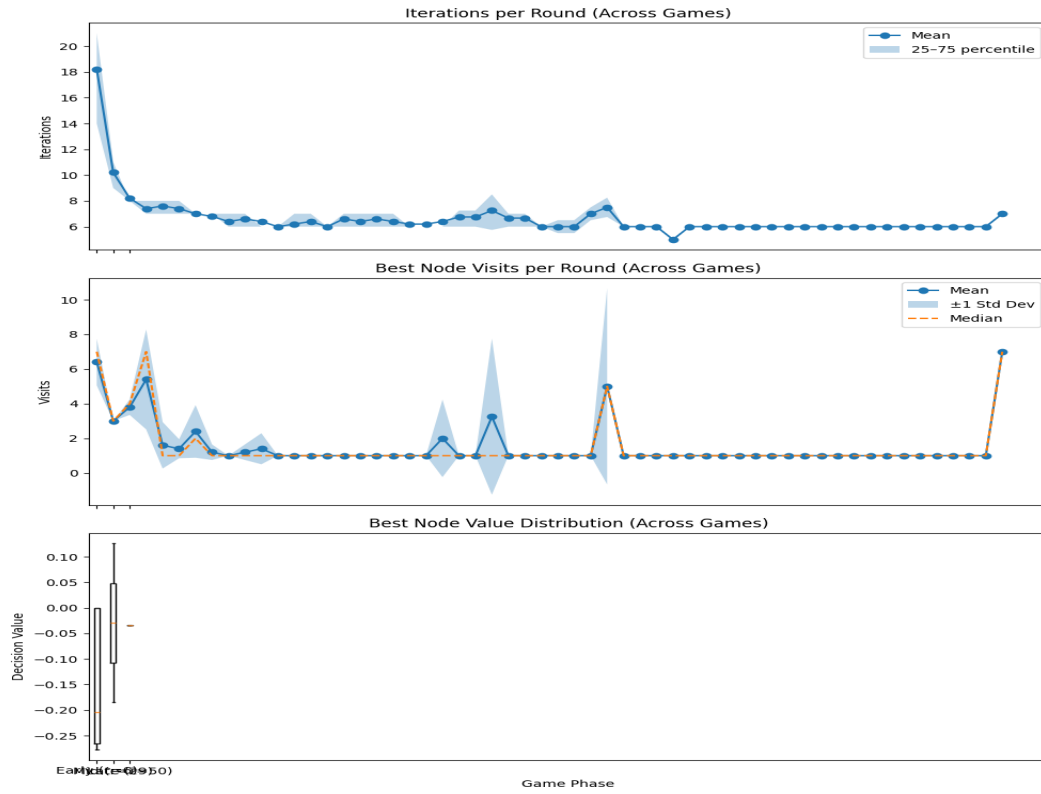
Avg tree depth per round (game-averaged):

Mean: 9.49

Min: 8.666666666666666

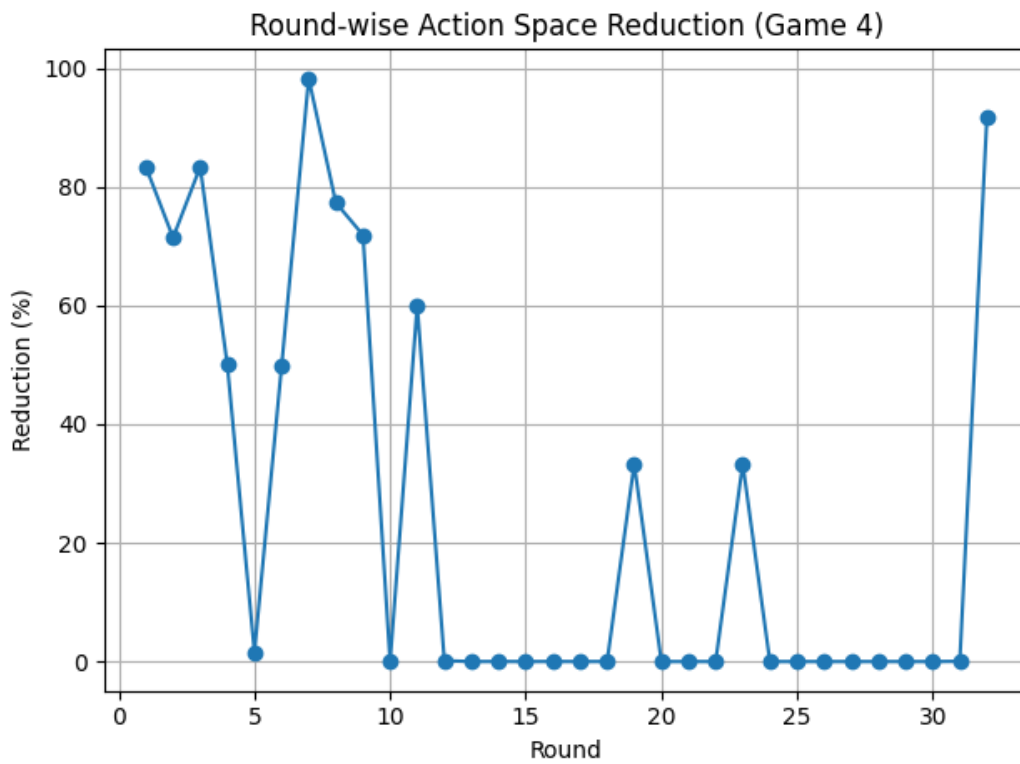
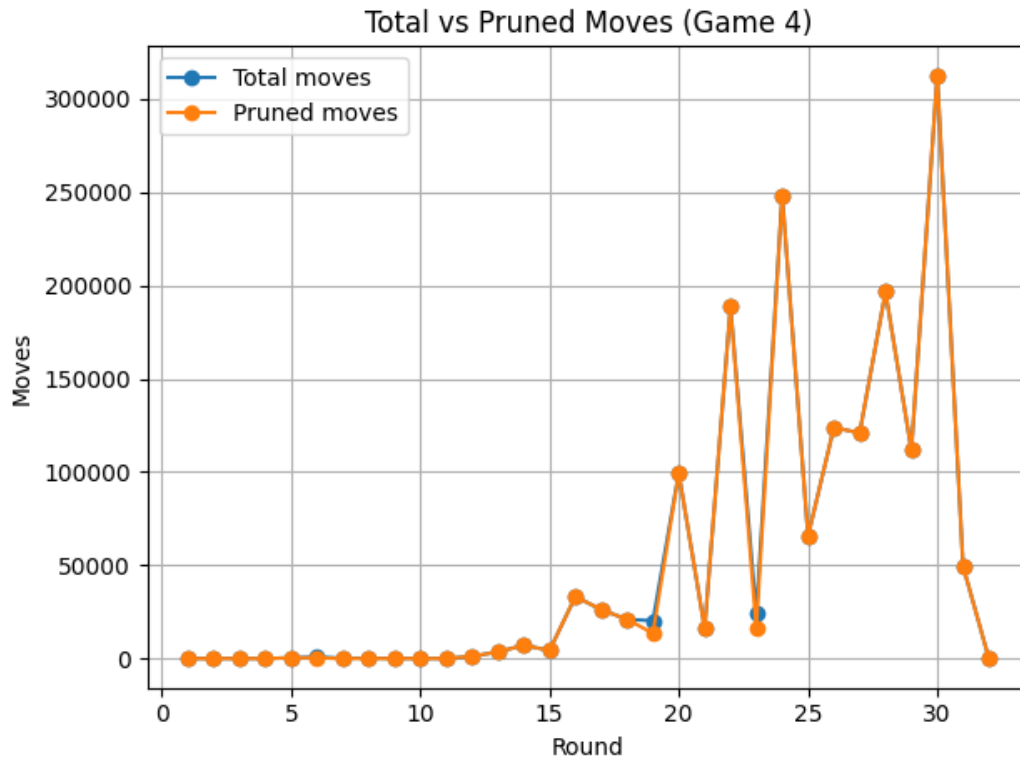
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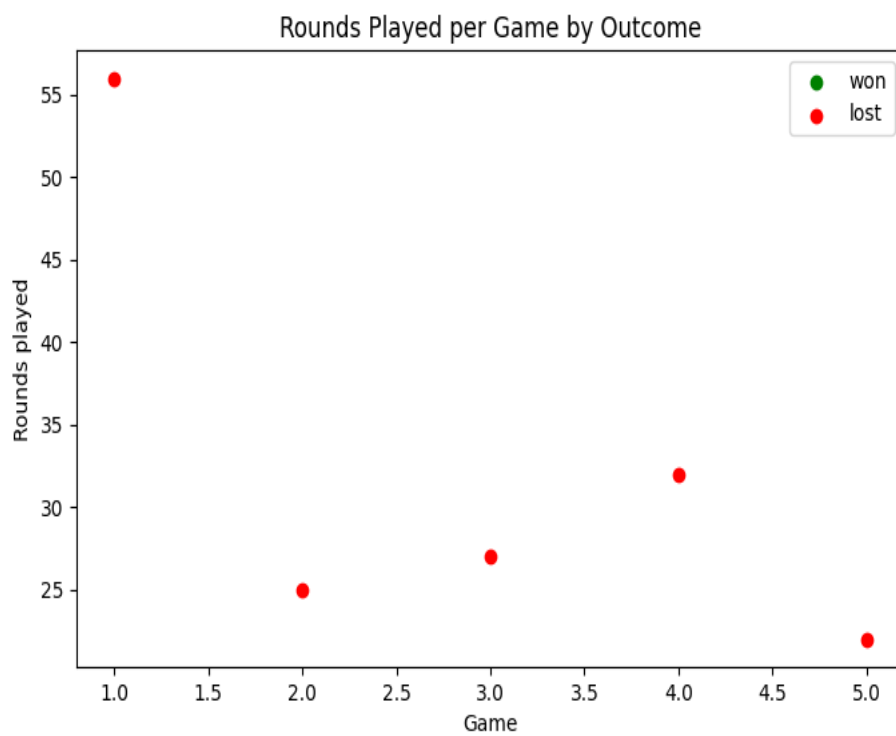
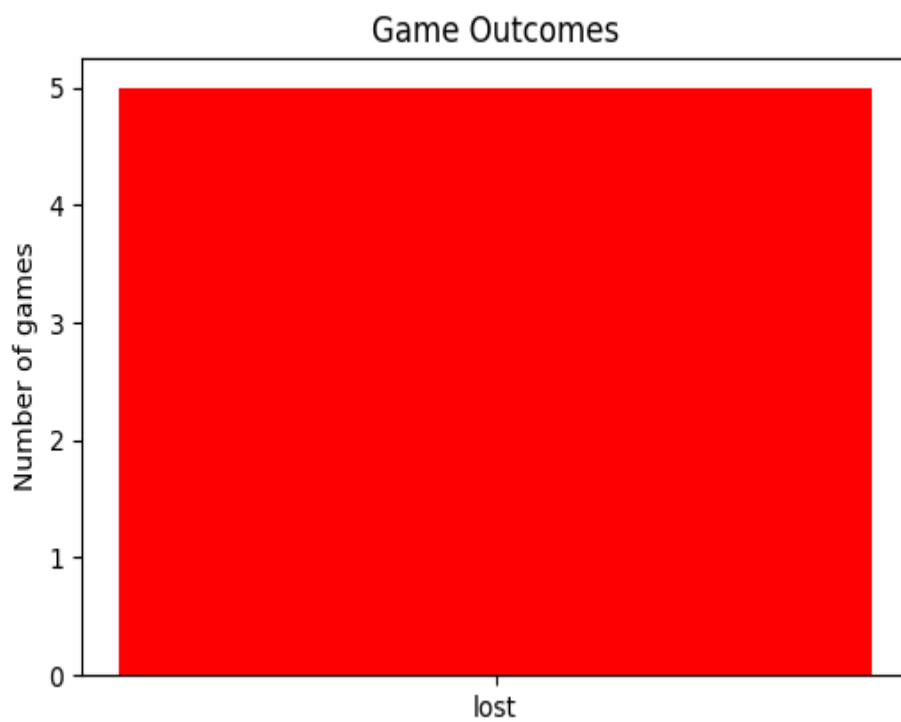


Action space reduction summary

game	rounds	avg_round_reduction	avg_round_reduction_percent
1	56.0	0.164013	16.401293
2	25.0	0.203274	20.327393
3	27.0	0.217372	21.737219
4	32.0	0.251708	25.170838
5	22.0	0.209946	20.994625
OVERALL	32.4	0.209263	20.926273



Game outcomes

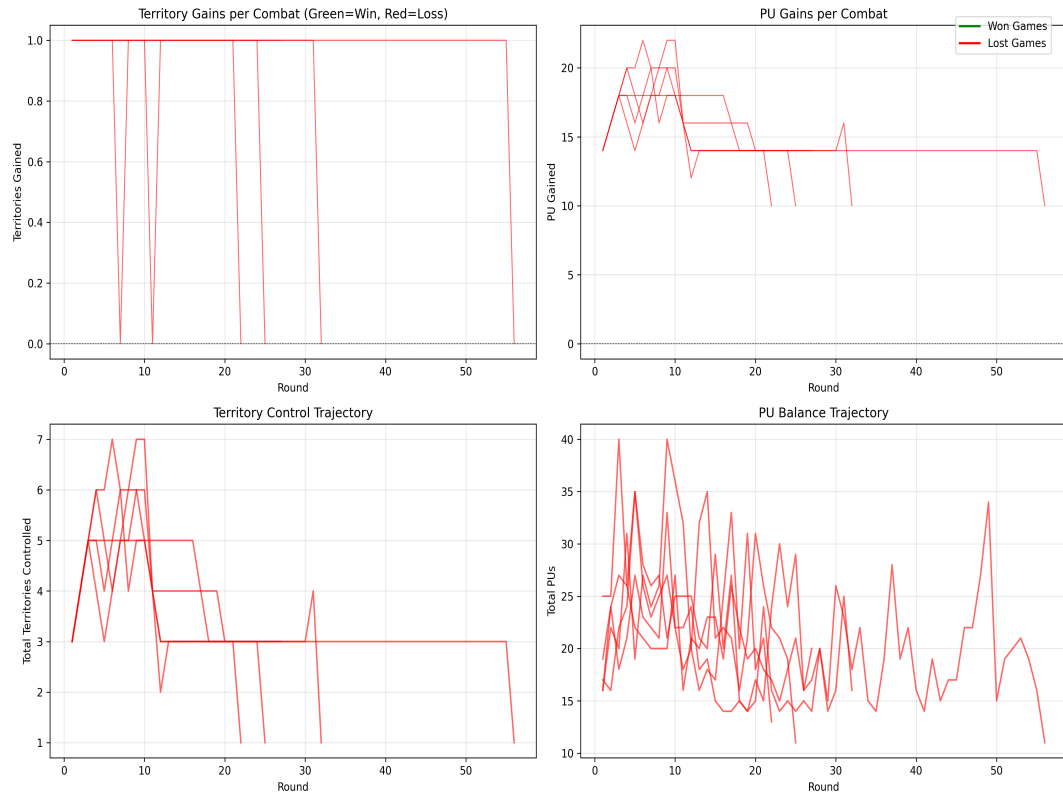


Combat move quality throughout the games

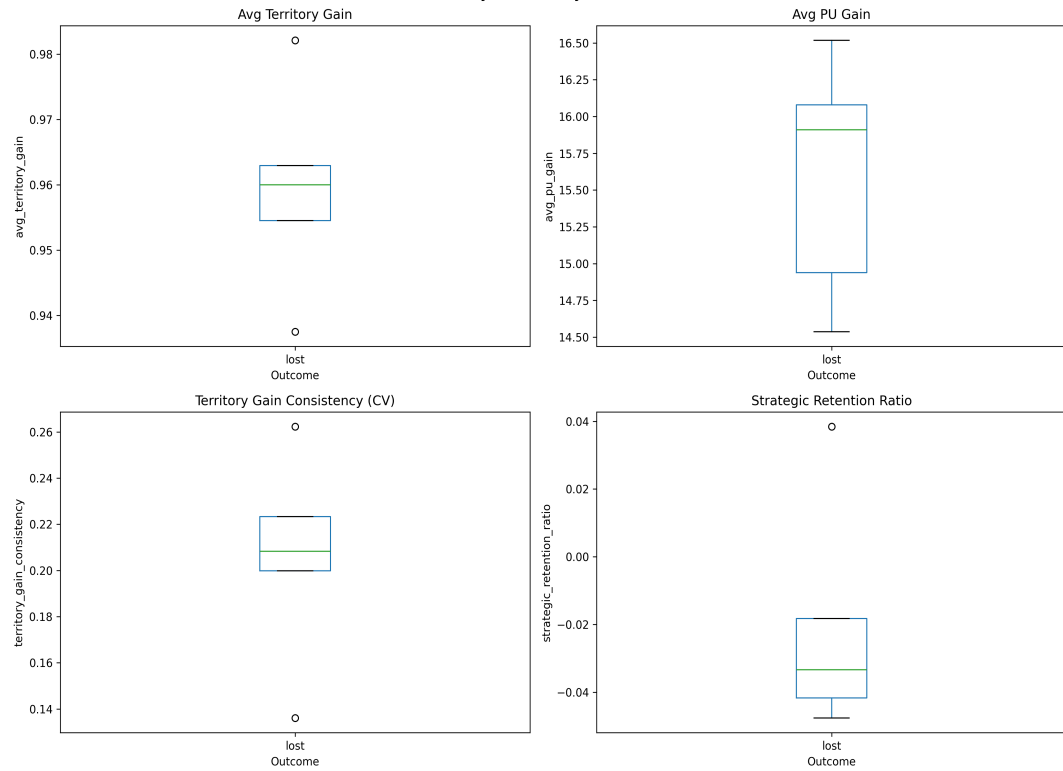
=== Combat Quality Metric Averages ===

Outcome: LOST

Avg Territory Gain: 0.959
Avg PU Gain: 15.596
Territory Gain Consistency: 0.206
Strategic Retention Ratio: -0.020



Combat Quality Metrics by Game Outcome



Profiling of MCTS algorithm

